



NETWORK DESIGN TEAM

(325)

FINAL

STATE 2021

Description

Analyze existing and planned business environments and develop a strategy for the implementation of a network infrastructure that addresses the business needs of the scenario provided.

Topic

Secure Medical Solutions is a contractor for several of the largest medical device and drug manufacturers throughout the United States. Being a reputable solution that provides services to high stake's medical companies with various trade secrets, security is high priority when it comes to data storage and retention and transmission. According to intellectual property rights, patent law, and the storage and transmission of personal medical information, Secure Medical Solutions' employees must follow strict guidelines at all times. Due to the novel coronavirus (Covid-19), multiple new manufacturers have retained Secure Medical Solutions to protect the development of vaccines candidates.

This rapid expansion has resulted in Secure Medical Solutions deciding to expand their network infrastructure.

One of the reasons Secure Medical Solutions has decided to do a redesign of their current network infrastructure is that several locations are being added outside the region where they are currently headquartered. The firm's current headquarters is located in Minneapolis, Minnesota, a satellite office exists in San Antonio, Texas and new offices are being created in Atlanta, Georgia and Salt Lake City, Utah.

Headquarters: Minneapolis, Minnesota

Secure Medical Solutions headquarters is a four (4) floor building in the historic district of Minneapolis, Minnesota. The first floor contains the following departments: reception and human resources. The second floor has the following departments: business marketing, customer support, and accounting. The third floor houses the IT department and secure datacenter. The fourth floor contains the executive offices and several conference rooms, each equipped with video conferencing equipment.

The first floor contains seventeen (17) workstations: two (2) of the workstations are assigned to reception, fifteen (15) are assigned to human resources. The second floor contains fifty (50) workstations: ten (10) are assigned to marketing, thirty (30) are assigned to customer support, and six (6) are assigned to accounting. All workstations on the first and second floors are connected per floor to an intermediate distribution frame (IDF) that contains a patch panel and a 48 port 10/100/1000 Mbps switch with a 10 Gbps uplink. From this switch, a 10 gigabit multi-mode fiber line runs up to the master distribution frame (MDF) in the third-floor datacenter.

The datacenter on the third floor contains 40 low-density racks, a majority of which contain virtual server hosts. A large storage area network (SAN) provides storage for the virtual servers.

hosted on the virtual. Secure Medical Solutions is requesting an additional secure backup solution for redundancy.

The fourth floor contains eight (8) executive offices with workstations, two (2) and all are connected to a 10/100/1000 Mbps switch, along with the three video conference rooms, which is then routed down to the datacenter on the third floor.

Satellite Office: San Antonio, Texas

The San Antonio office contains the following departments: legal and efficacy data review. This office is located in the medical research district of San Antonio and all employees at this office will need secure access to virtual servers or storage located at the headquarters.

Forty (40) workstations reside in the office and are connected to an encrypted local file storage server. Secure Medical Solutions is requesting a fast and secure cloud-based solution that will be used for redundancy.

New Satellite Offices: Atlanta, Georgia and Salt Lake City, Utah

Each of the new offices will be rented from other established companies and will take up two floors of each of the respective buildings. The new satellite offices each will consist of forty (40) cubicles, twenty (20) per floor and will need high speed access to the secure storage servers at the headquarters. Employees located at these offices will be assigned company laptop computers and no workstations will be permanently located at the locations. Since, the office building is in a shared space, employees will need to work within a secure connection and company assets must be properly managed. Additionally, the company has requested IP security cameras and biometric security measures to be present at each satellite office.

Customer's needs:

- Provide a fast, secure, and efficient network design for the two new offices and a method for connection to the main office.
- Provide a solution for the segmentation of the network in the main office as all machines are currently connected to the same network.
- Provide a solution for maintaining secure network connections in the shared spaces of the satellite offices, including security cameras and biometric security measures.
- Recommend an effective design for the subnetting of the entire network.
- Devise a solution cloud-based redundancy for the San Antonio location.
- Provide a solution for company asset management.
- Suggest a strategy for improving the current security of the network as well as security in the new offices.
- Develop a network design that is modern, secure, reliable, and compliant to intellectual property and trademark storage guidelines.
- Design a strategy for effective and secure communication amongst remote employees.

Modification for State Final:

Secure Medical Solutions has requested that teleconferencing rooms be added to each of the new satellite offices in Atlanta, Georgia and Salt Lake City, Utah. Your team is tasked with presenting a brief of the technology for the teleconference room. Items to include are, but not limited to: wireless projection and interactive display technology, secure video/audio delivery service.